

Security Analyst

Summary

- Monitor activity taking place on the devices and network of an organization.
- Investigate unusual or suspicious activity ie: strange logins.
- Work with other teams in a their system to improve the organization's defenses.

Responsibilities, Skills & Tools

Security analysts serve as the first responders to digital threats.

Tasks and duties include:

Continuous Monitoring: Observing SIEM dashboards 24/7/365 for anomalous activity.

Alert Triage: Separating "noise" (false positives) from genuine attacks (true positives).

Incident Investigation: Using endpoint telemetry (EDR) and network logs to reconstruct attack timelines.

Containment & Eradication: Taking technical actions like isolating infected hosts, revoking credentials, or blocking malicious IPs.

Sample deliverables

- Python-Based Security "Swiss Army Knife": Custom scripts for automated reconnaissance, such as a tool that finds subdomains or scans for open ports using the Nmap library.
- Cryptographic Password Manager: A password manager in Python that uses a cryptography library to encrypt and securely store credentials, demonstrating your understanding of data persistence and security principles.

- - Automated Data Quality & QA Pipeline: An ETL (extract, transform, load.) pipeline that takes raw, "messy" data (like inconsistent address lists or incident logs), geocodes it, and scores it for accuracy to reduce manual backlogs.
- - IoT Security Vulnerability Lab: Set up a home lab to test connected devices. Analyze firmware for hardcoded credentials or unencrypted communication protocols, then document how you would harden the device.